

# ECEN 215, Spring 2024

## Electrical Circuit Theory

### Lecture 6: Capacitor, Inductor, Transient

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Ramy Rady

Dept. of Electrical and Computer Engineering

Texas A&M University

[ramyrady@tamu.edu](mailto:ramyrady@tamu.edu)



# OUTLINE

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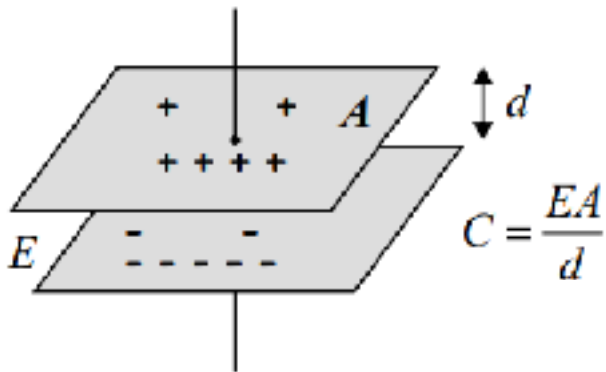


- Capacitors
- Inductors
- Transient circuit analysis
  - Natural/Step Response R-C/R-L Circuits
- Next Lecture

# Capacitor



- Capacitor is a passive electrical component consists of two electrical conductors separated by an insulator and the capacitance is related to area of the conductors and the spacing between them, where  $\epsilon$  is the dielectric primitivity of the insulator
- Capacitance (C) is measured in Farad (F) named after Michael Farady

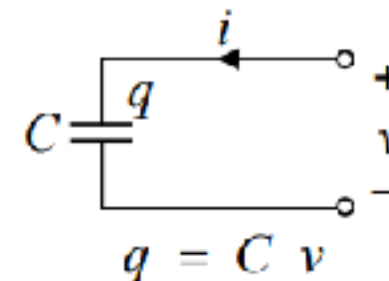


Michael Farady

# Capacitor



- Capacitors store energy in the form of an electric field between two charged plates (sort of like it is “storing voltage”)



## Energy Stored in a Capacitor

**Work** done by a battery/power supply in "charging" a capacitor is stored as **electrical potential energy** in an **electric field** which exists between the charged capacitor plates.

This **electrical potential energy** is released when the capacitor is **discharged**, e.g., by connecting both plates of the capacitor to a light bulb.

$$E = \frac{1}{2} Q V = \frac{1}{2} C V^2 = \frac{1}{2} \frac{Q^2}{C}$$

$E$  = energy stored in capacitor/ joules (J)

$Q$  = charge stored on capacitor plates/ coulombs (C)

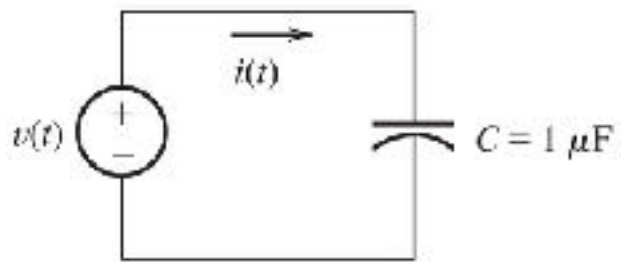
$V$  = potential difference between capacitor plates/ volts (V)

$C$  = capacitance of capacitor/ farads (F)

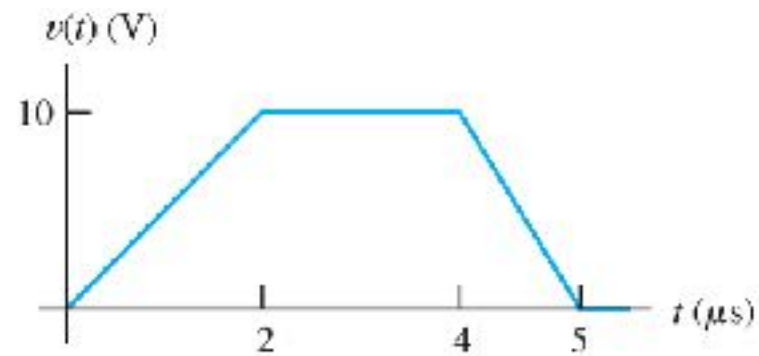
$$\begin{aligned} i &= \frac{dq}{dt} \\ &= \frac{d(Cv)}{dt} \\ &= C \frac{dv}{dt} \end{aligned}$$

$$\left[ E = \frac{1}{2} C v^2 \right]$$

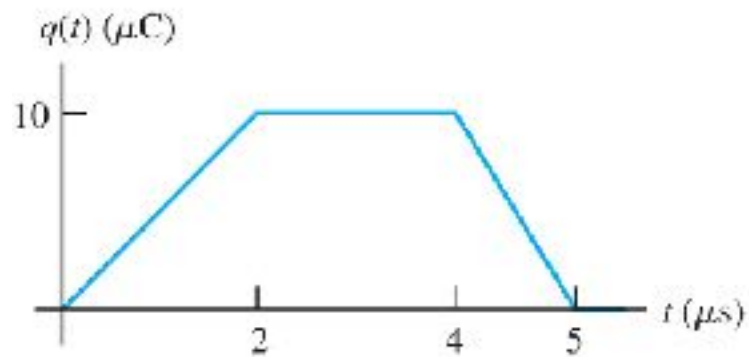
# Example 1) Capacitor Current



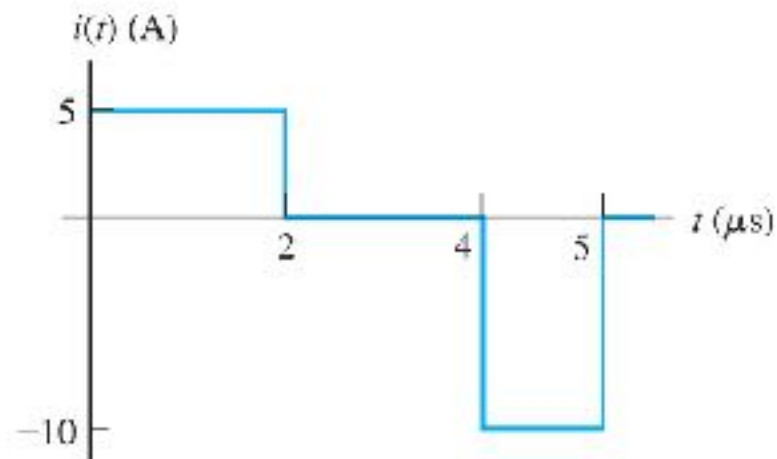
(a)



(b)



(c)

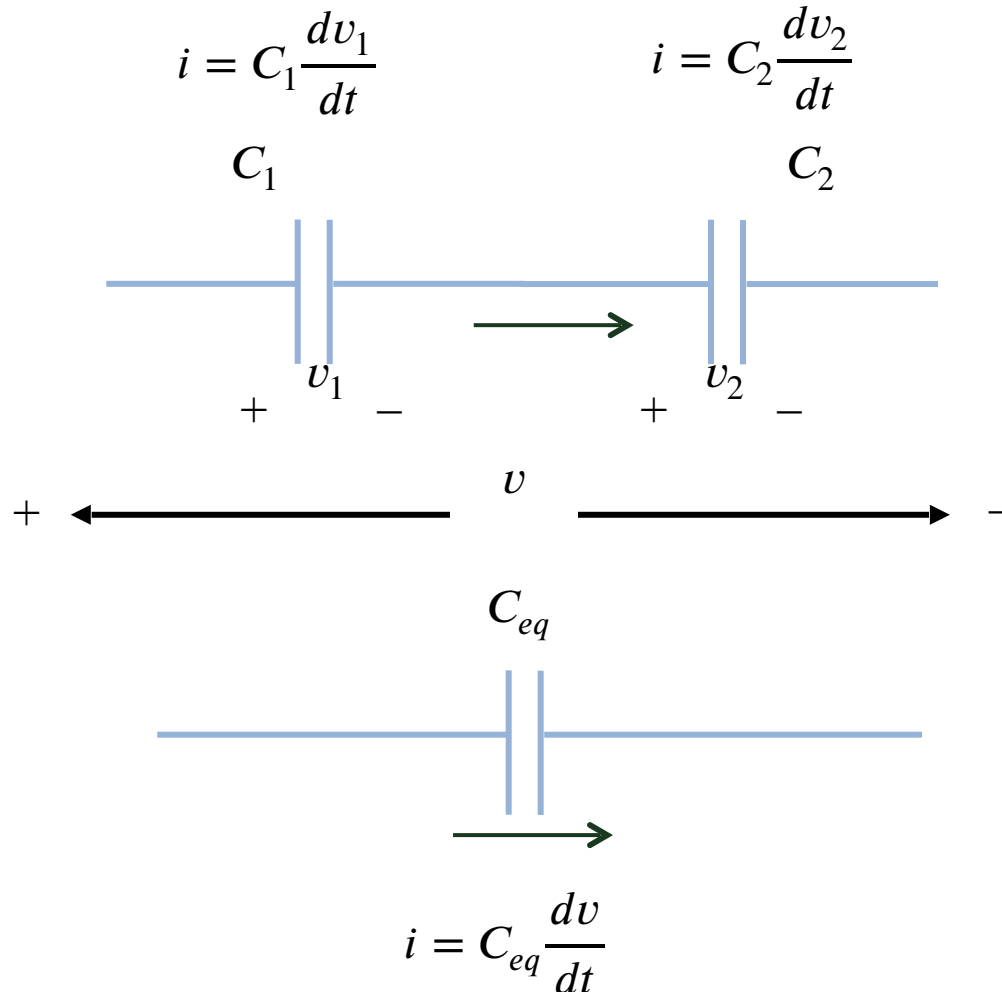


(d)

# Capacitors in Series



- Capacitors in series combine like resistors in parallel.



Using KVL:  $v = v_1 + v_2$

Differentiating:  $\frac{dv}{dt} = \frac{dv_1}{dt} + \frac{dv_2}{dt}$

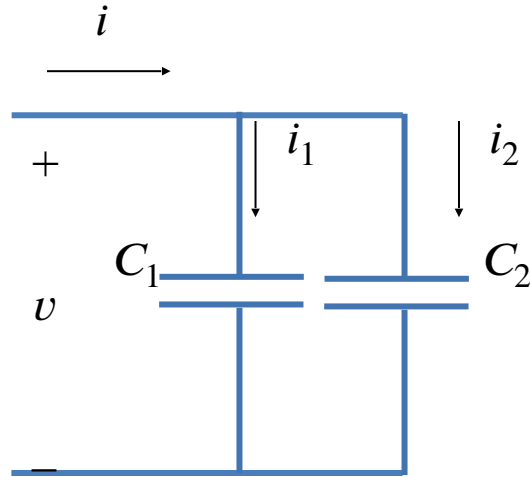
Substituting:  $\frac{i}{C_{eq}} = \frac{i}{C_1} + \frac{i}{C_2}$

$$\boxed{\frac{1}{C_{eq}} = \frac{1}{C_1} + \frac{1}{C_2}}$$

# Capacitors in Parallel



- Capacitors in parallel combine like resistors in series.



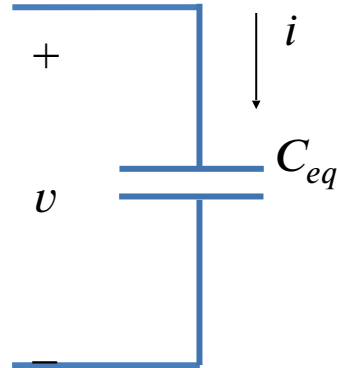
$$i_1 = C_1 \frac{dv}{dt}$$

$$i_2 = C_2 \frac{dv}{dt}$$

Using KCL:  $i = i_1 + i_2$

Substituting:

$$\begin{aligned} C_{eq} \frac{dv}{dt} &= C_1 \frac{dv}{dt} + C_2 \frac{dv}{dt} \\ &= (C_1 + C_2) \frac{dv}{dt} \end{aligned}$$



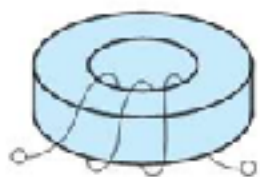
$$i = C_{eq} \frac{dv}{dt}$$

$$C_{eq} = C_1 + C_2$$

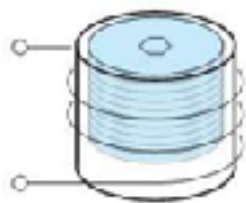
# Inductor



- An inductor, also called a coil, is a passive two-terminal electrical component that stores energy in a magnetic field. It typically consists of an insulated wire wound into a coil.
- Inductance ( $L$ ) is measured in Henry (H) named after Joseph Henry



(a) Toroidal inductor



(b) Coil with an iron-oxide slug that can be screwed in or out to adjust the inductance

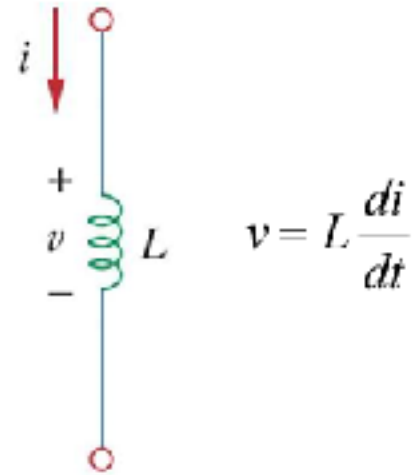
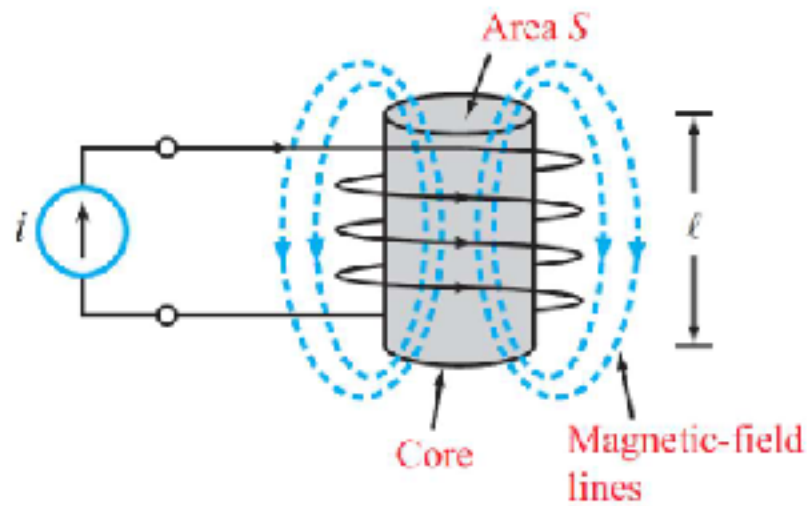


Joseph Henry

# Inductor



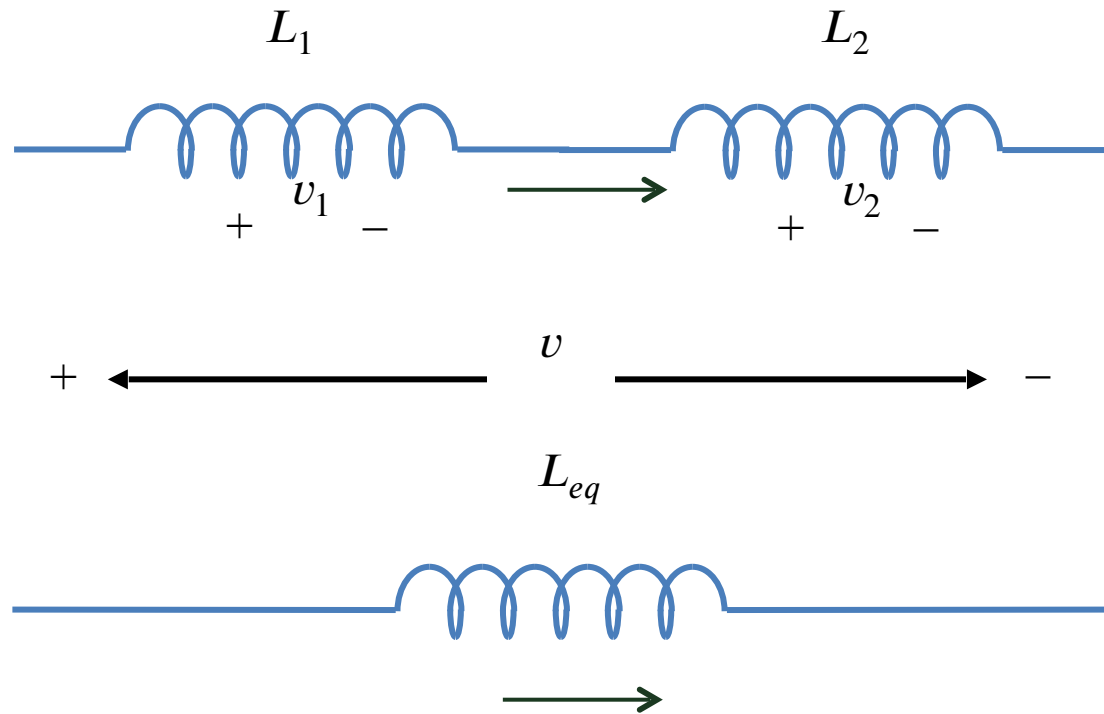
- Inductors store energy in the form of an magnetic field (sort of like it is “storing current”)



# Inductors in Series



- Inductors combine in series exactly like resistors.



$$v_1 = L_1 \frac{di}{dt} \quad v_2 = L_2 \frac{di}{dt}$$

$$v = v_1 + v_2 = (L_1 + L_2) \frac{di}{dt}$$

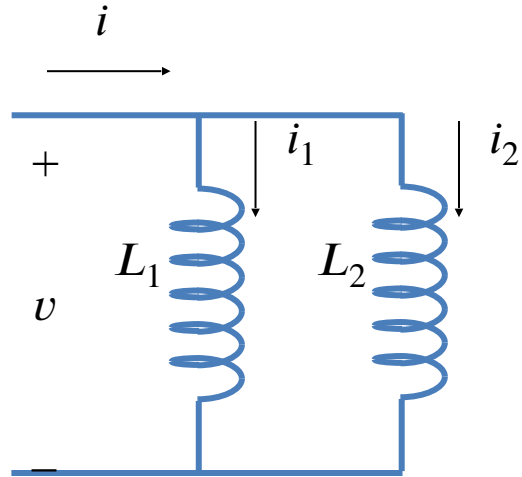
$$v = L_{eq} \frac{di}{dt}$$

$$L_{eq} = L_1 + L_2$$

# Inductors in Parallel



- Inductors combine in parallel exactly like resistors.

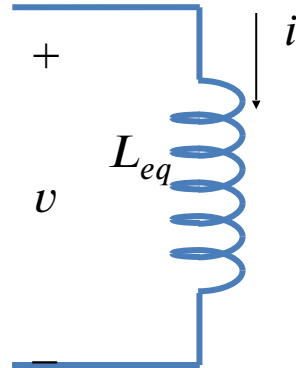


$$v = L_1 \frac{di_1}{dt}$$

$$v = L_2 \frac{di_2}{dt}$$

Using KCL:  $i = i_1 + i_2$

Differentiating:  $\frac{di}{dt} = \frac{di_1}{dt} + \frac{di_2}{dt}$



$$v = L_{eq} \frac{di}{dt}$$

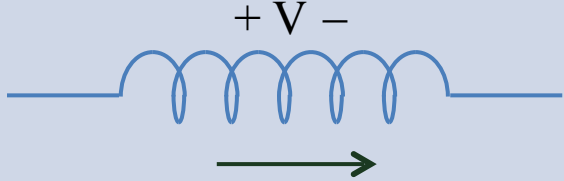
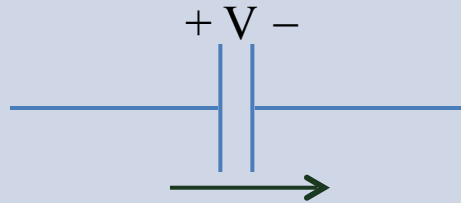
Substituting:  $\frac{v}{L_{eq}} = \frac{v}{L_1} + \frac{v}{L_2}$

$$\boxed{\frac{1}{L_{eq}} = \frac{1}{L_1} + \frac{1}{L_2}}$$

# Inductors vs. Capacitors



- Inductors and capacitors are passive devices which store (and release) energy (energy is not created/generated by a passive device).

|                                      | Inductor                                                                           | Capacitor                                                                           |
|--------------------------------------|------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------|
| <b>Circuit Symbol</b>                |  |  |
| <b>Fundamental I/ V relationship</b> |                                                                                    |                                                                                     |
| <b>Units</b>                         | L=inductance (Henrys)                                                              | C=capacitance (Farads)                                                              |



# Properties of Inductors and Capacitors

- From the fundamental I/V relationship we can infer several properties of inductors and capacitors.

|                                                              | Inductor                                                                                                                                                        | Capacitor                                                                                                                                                       |
|--------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Fundamental I/V relationship</b>                          |                                                                                                                                                                 |                                                                                                                                                                 |
| <b>Instantaneous changes</b>                                 | The current in an inductor cannot change instantaneously. To do so would cause an infinite voltage to appear.                                                   | The voltage across a capacitor cannot change instantaneously. To do so would cause an infinite current to flow.                                                 |
| <b>Steady State (when things are not changing with time)</b> | When a circuit with an inductor reaches steady state, the voltage across the inductor will be zero.<br>→ An inductor acts like a short circuit in steady state. | When a circuit with a capacitor reaches steady state, the current through the capacitor will be zero. → An capacitor acts like an open circuit in steady state. |



# Inverse to Fundamental I–V Relationships

- From the fundamental I–V relationships we can also derive inverse relationships.

## Inductors

$$v = L \frac{di}{dt}$$

$$\int_{t_o}^t v(u) du = L \int_{t_o}^t \frac{di}{du} du$$

$$\int_{t_o}^t v(u) du = L(i(t) - i(t_o))$$

$$i(t) = i(t_o) + \frac{1}{L} \int_{t_o}^t v(u) du$$

## Capacitors

$$i = C \frac{dv}{dt}$$

$$\int_{t_o}^t i(u) du = C \int_{t_o}^t \frac{dv}{du} du$$

$$\int_{t_o}^t i(u) du = C(v(t) - v(t_o))$$

$$v(t) = v(t_o) + \frac{1}{C} \int_{t_o}^t i(u) du$$

# Power and Energy in Inductors and Capacitors



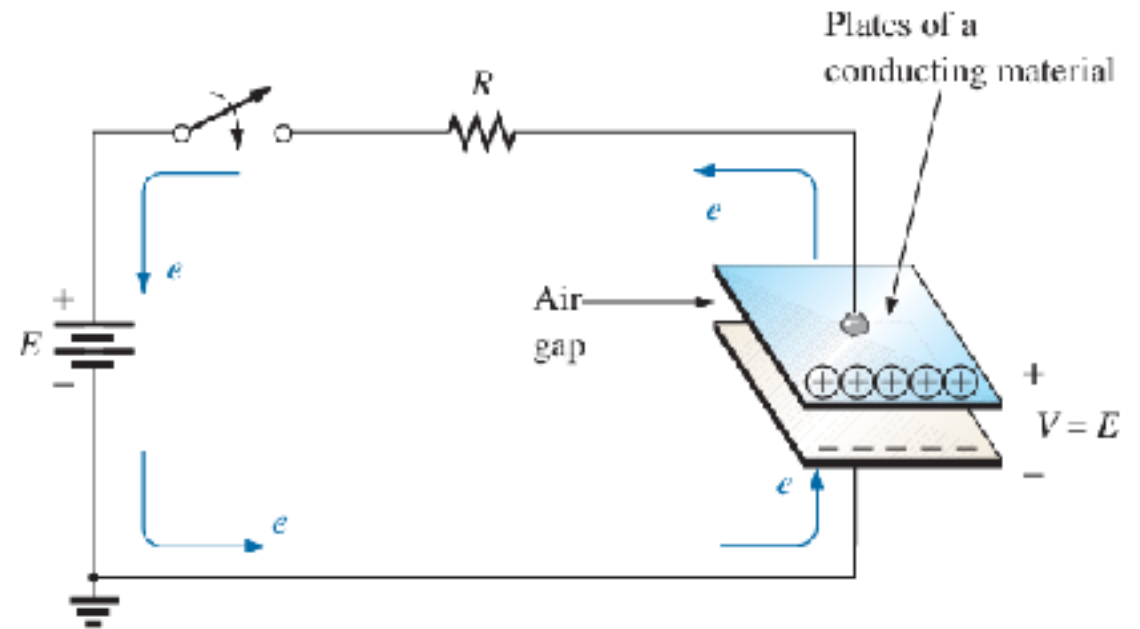
- From the fundamental I–V relationship we can infer several properties of inductors and capacitors.

|                                     | Inductor | Capacitor |
|-------------------------------------|----------|-----------|
| <b>Fundamental I–V relationship</b> |          |           |
| <b>Power</b>                        |          |           |
| <b>Energy</b>                       |          |           |

# Capacitor Circuits in Transient: How it works



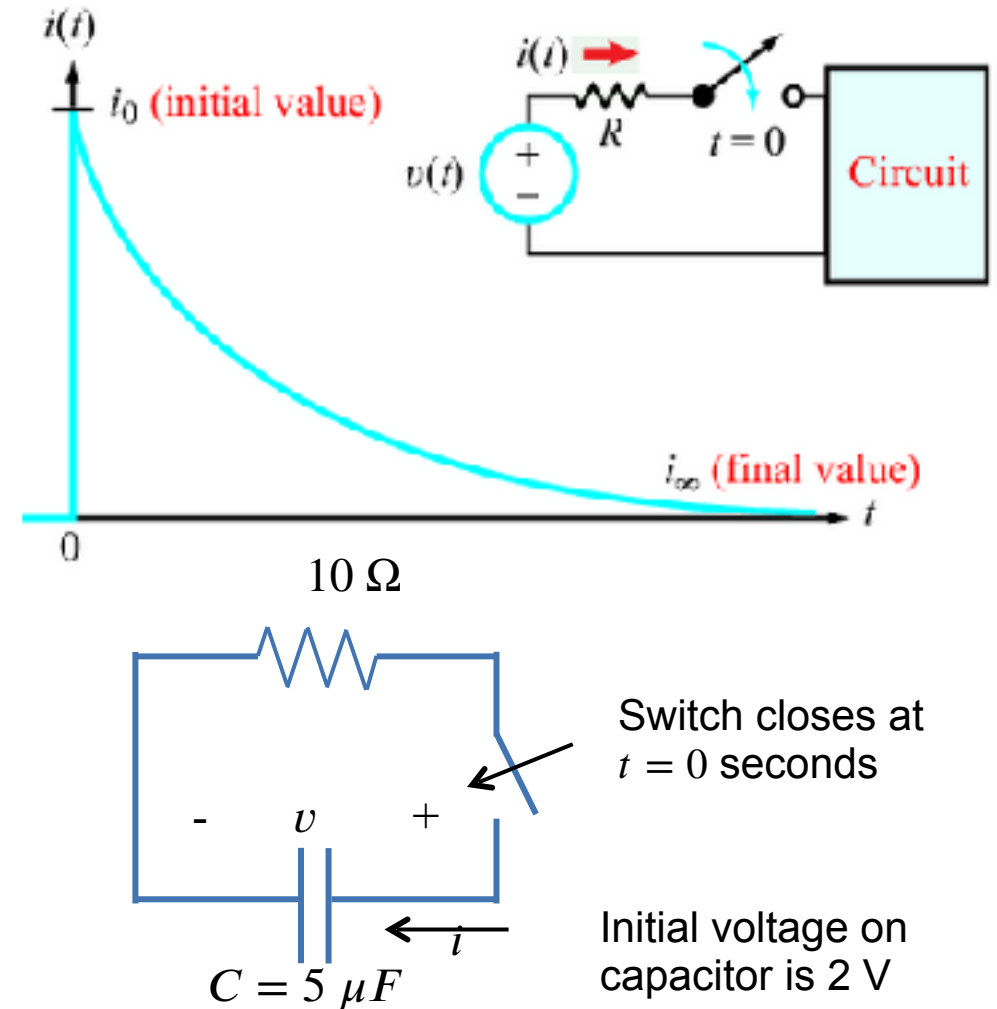
- Now, we will look at how circuits change over time (Transient)
- Capacitor circuit when switch is closed
  - electrons are drawn from the upper plate through
  - the resistor to the positive terminal of the battery
  - Surge in current limited in magnitude by  $R$
  - a net positive charge on the top plate.
  - until the potential difference  $V = E$



# Transient Circuit Analysis



- We'll solve these
- (1) steady-state before the switch  
( $t < 0$ )  $\rightarrow$  C: O.C. and L is Sh.C
- (2) instantaneously after the switch  
( $t=0$ )
- (3) steady-state after the switch  
( $t > 0$ )  $\rightarrow$  C: O.C. and L is Sh.C



# Example 2 discharging RC Circuit



1. Steady-state before the switch:

$$v = 2 \text{ V}, i = 0 \text{ A}$$

$$\text{Energy stored in the capacitor} = \frac{1}{2}v^2C = 10 \mu\text{J}$$

2. Instantaneously after the switch:

$$v = 2 \text{ V}$$

$$i = -\frac{2\text{V}}{10\Omega} = -0.2 \text{ A}$$

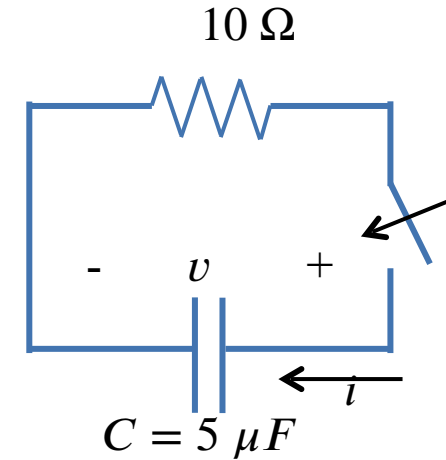
$$\text{Equation: } C \frac{dv(t)}{dt} = i(t) = -\frac{v(t)}{R}$$

3. In steady state after the switch:

$$i = 0 \text{ A}$$

$$v = 0 \text{ V}$$

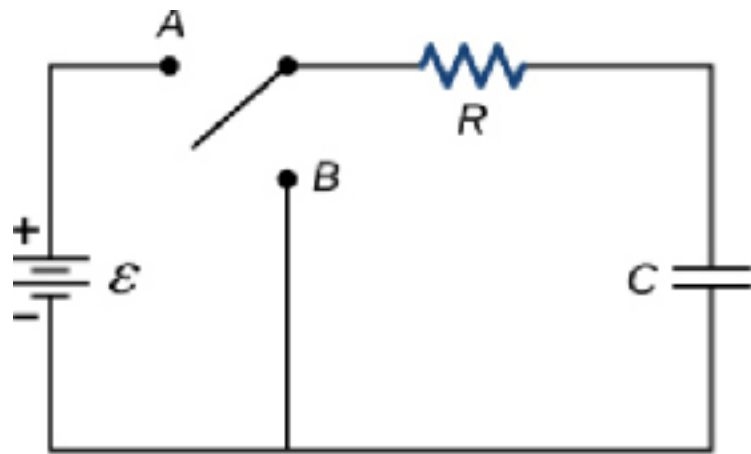
$$\text{Energy stored in the capacitor} = 0 \text{ J}$$



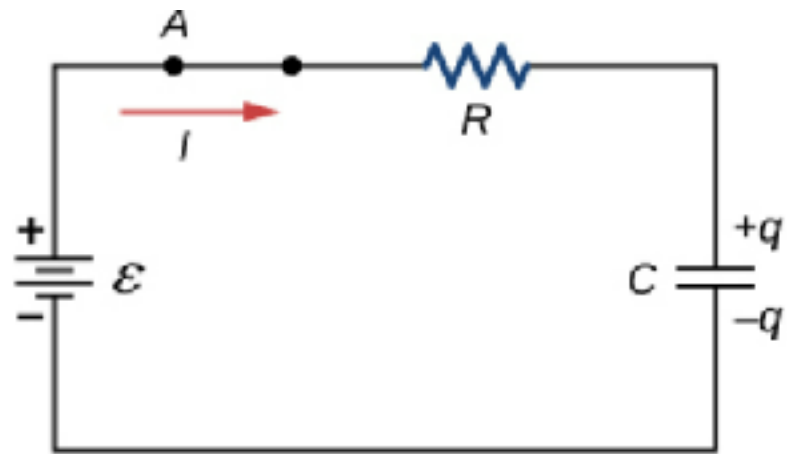
Switch closes at  
 $t = 0$  seconds

Initial voltage on  
capacitor is 2 V

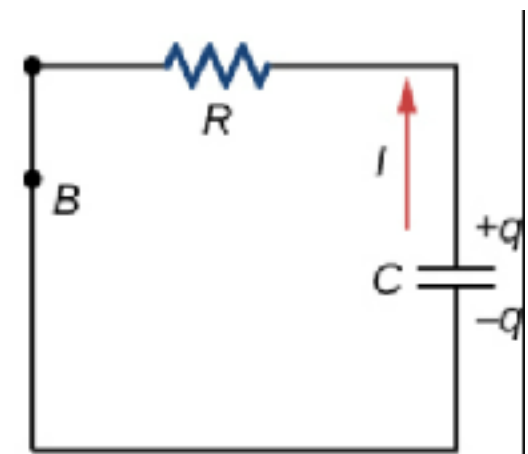
# Capacitor Charging and discharging



(a) Original circuit

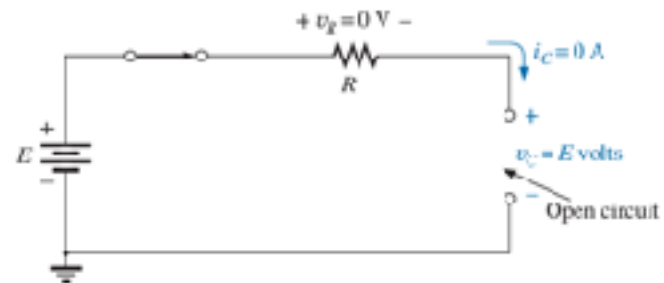


(b) Charging capacitor



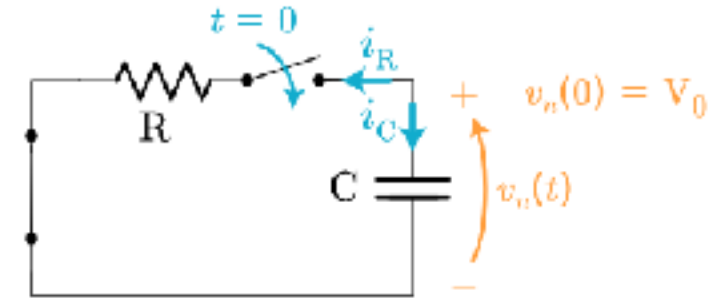
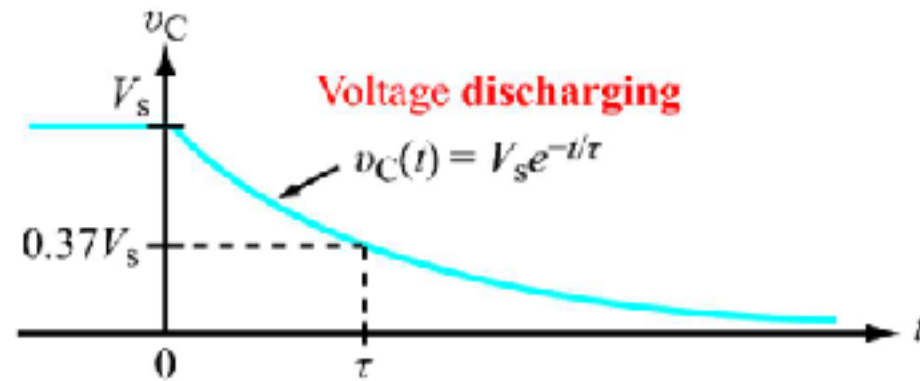
(b) Discharging capacitor

*A capacitor can be replaced by an open-circuit equivalent once the charging phase in a dc network has passed.*



*Demonstrating that a capacitor has the characteristics of an open circuit after the charging phase has passed.*

# Natural Response (discharging) capacitor

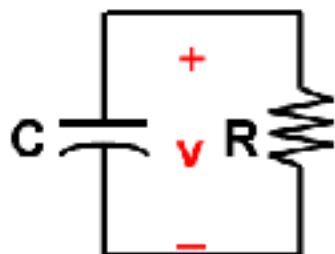


Time constant:  $\tau = RC$

# Natural Response RL and RC



RC Circuit



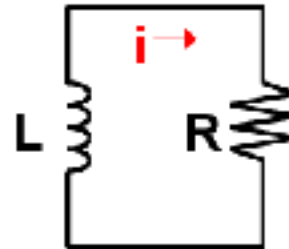
- Capacitor voltage cannot change instantaneously

$$v(0^-) = v(0^+)$$

$$v(t) = v(0)e^{-t/\tau}$$

- time constant  $\tau = RC$

RL Circuit



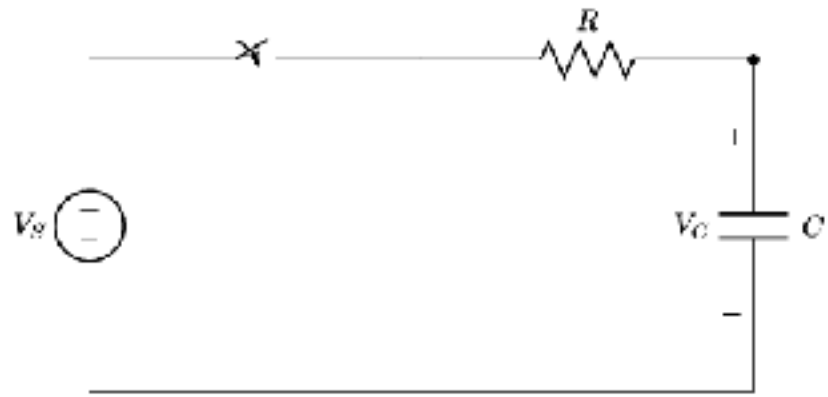
- Inductor current cannot change instantaneously

$$i(0^-) = i(0^+)$$

$$i(t) = i(0)e^{-t/\tau}$$

- time constant  $\tau = \frac{L}{R}$

# Capacitor Charging



$$\frac{V_S - V_C}{R} = i_C$$

$$\frac{V_S - V_C}{R} = C \frac{dV_C}{dt}$$

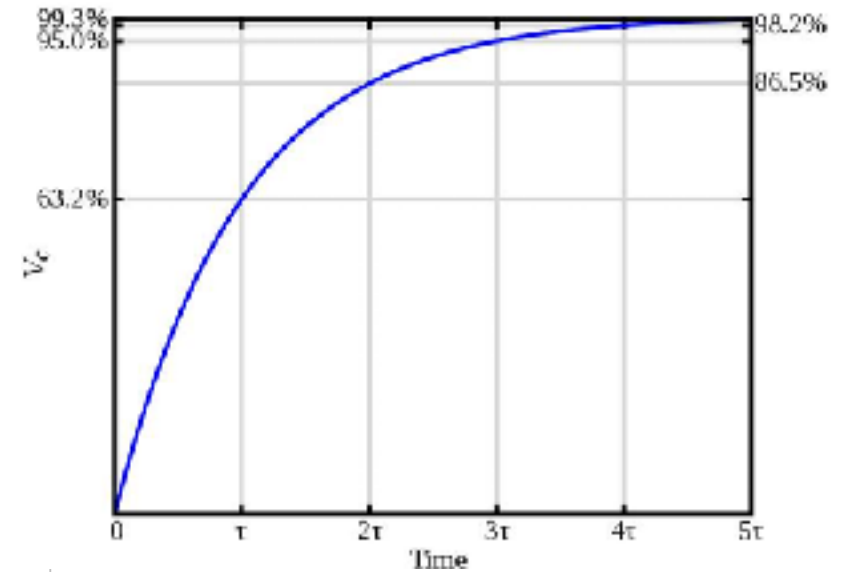
$$\frac{dt}{RC} = \frac{dV_C}{V_S - V_C}$$

$$\int_0^t \frac{1}{RC} dt = \int_0^t \frac{1}{V_S - V_C} dV_C$$

$$\frac{t}{RC} = -\ln(V_S - V_C) + C_1$$

$$C_2 e^{\frac{-t}{RC}} = V_S - V_C$$

$$V_C(t) = V_S(1 - e^{\frac{-t}{RC}})$$



# What You Need to Know Today



- Capacitors
  - Fundamental equation:  $i = C \frac{dv}{dt}$ , where  $C$  is capacitance in Farads
  - Add in parallel, combine in series the way resistors do in parallel
  - In steady-state, they act like an open circuit
  - Energy stored is  $Cv^2$
- Inductors
  - Fundamental equation:  $v = L \frac{di}{dt}$ , where  $L$  is inductance in Henries
  - Combine in series and parallel the same way as resistors
  - In steady-state, they act like a short circuit
  - Energy stored is  $Li^2$
- Transient circuit solutions
  - We are solving for  $v$  and  $i$  as a function of time,  $v(t)$  and  $i(t)$
  - Write the differential equations using KVL and KCL
  - Next class we will start to work on solving them

# Next Lecture

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- Phasor